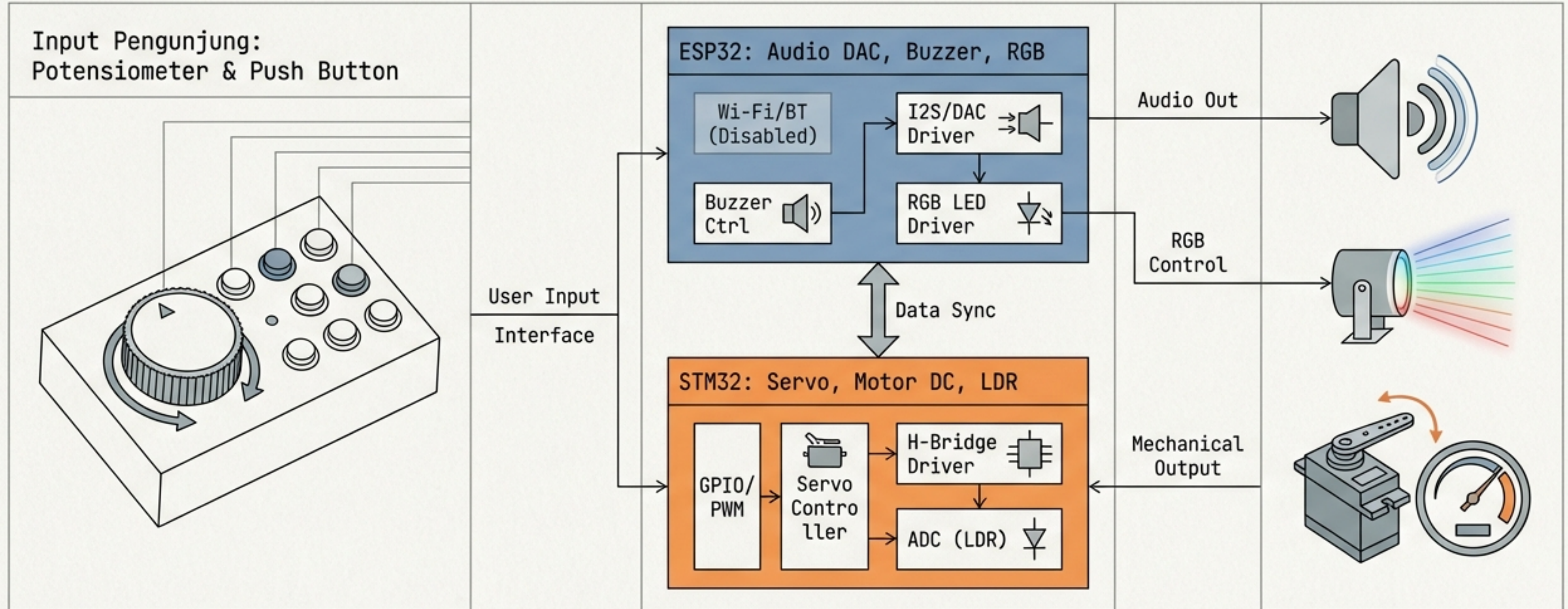


Museim Interactive Controller

Sistem Kontrol Audio & Cahaya Dual-MCU

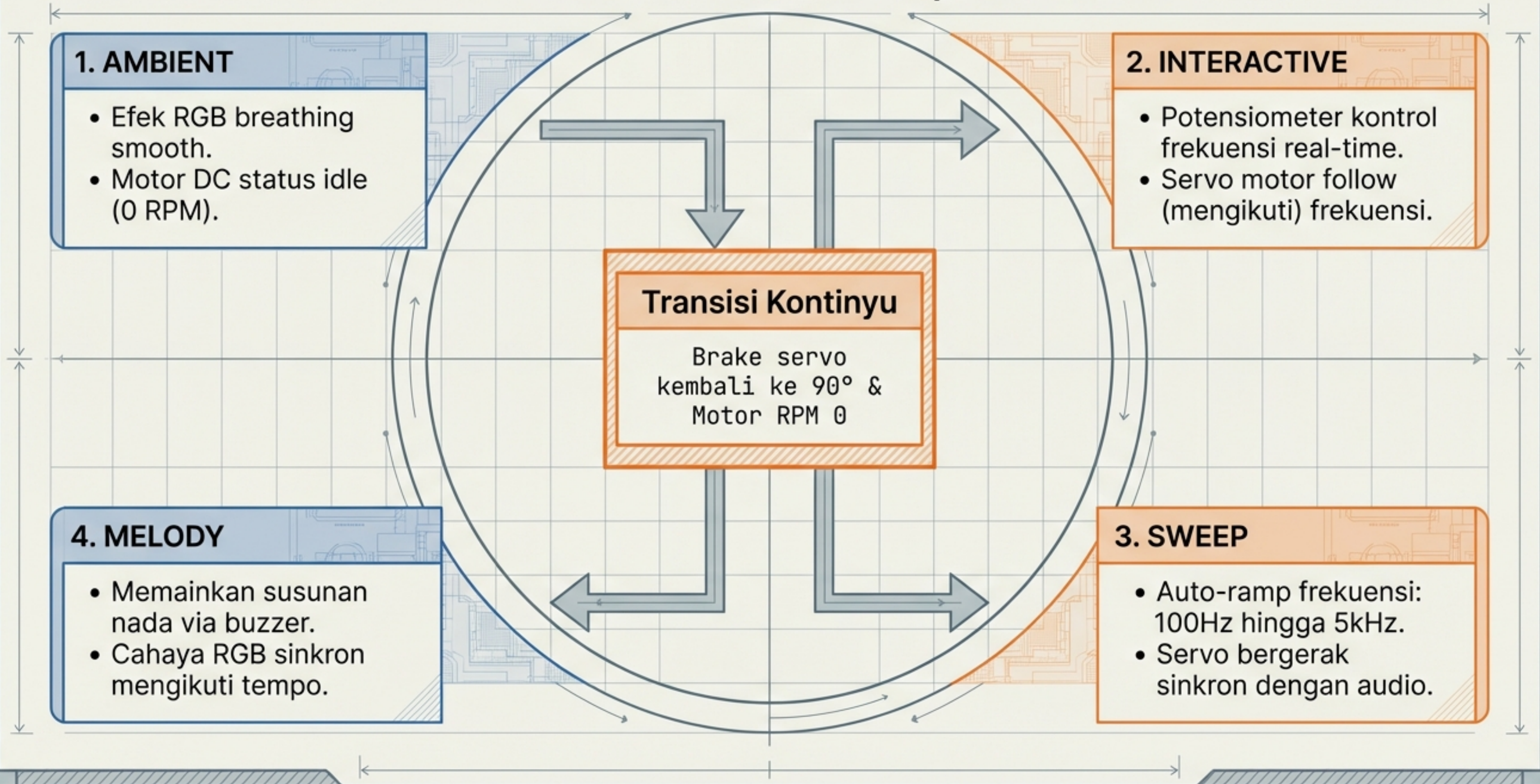


Waktu Development: 2 Minggu

Arsitektur: ESP32 & STM32F103

Protokol: UART Synchronous Swap

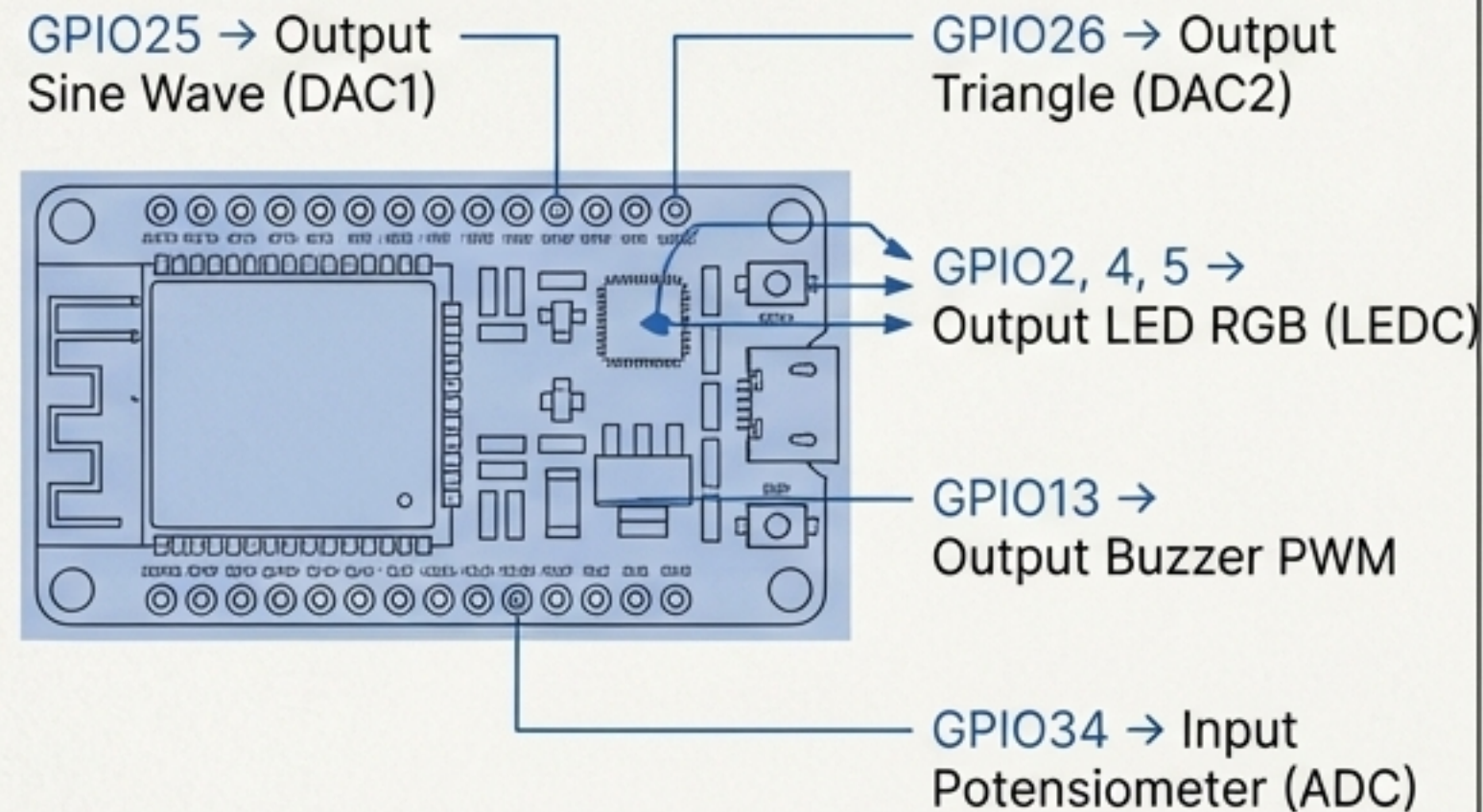
Finite State Machine: Mode Operasi Sistem



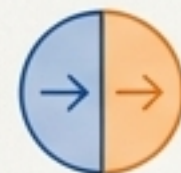
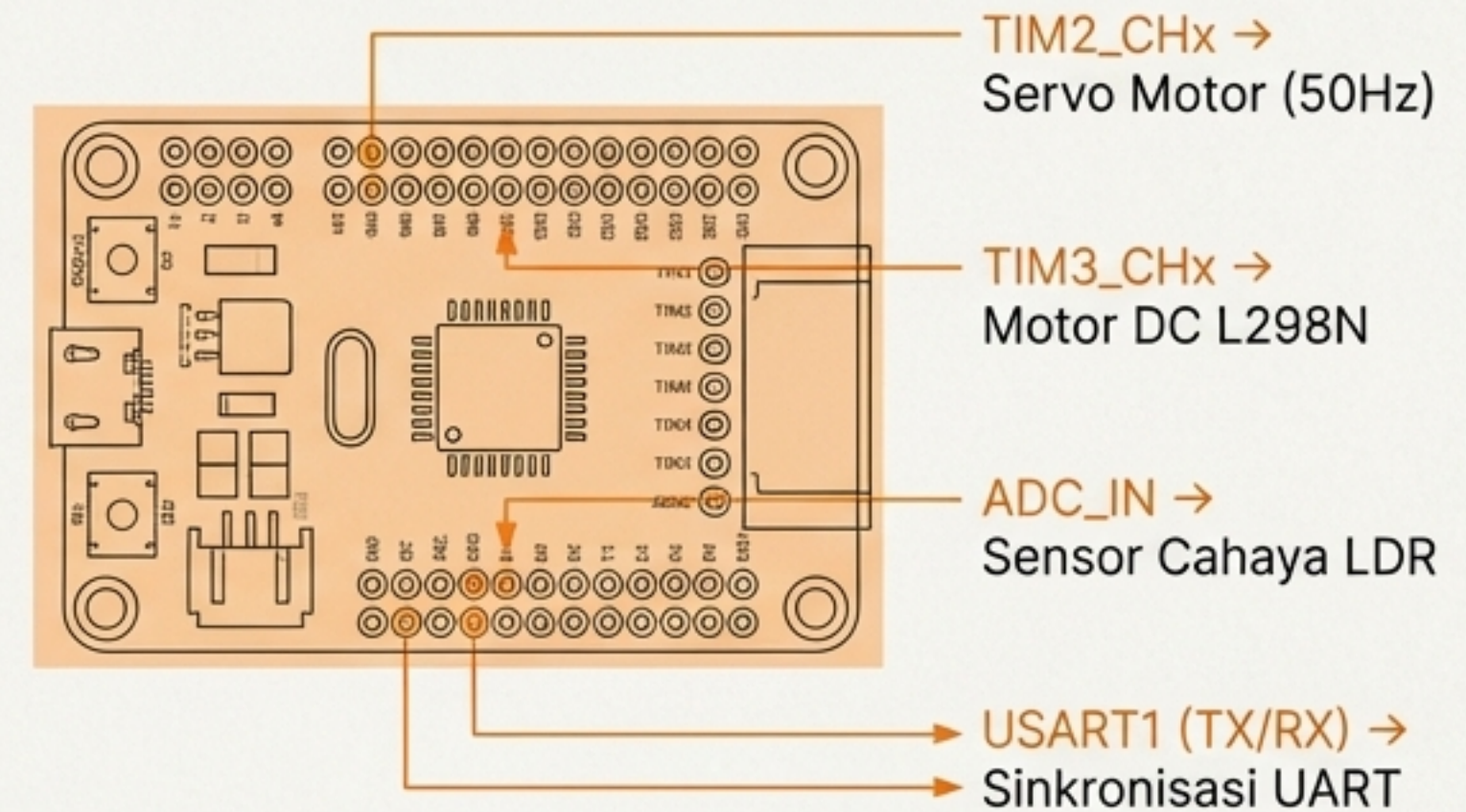
Spesifikasi Inter Hardware & Pemetaan Pin

Integrasi P01-P12: Sine, Triangle, Breathing, Servo, Motor, RGB, MeLody

ESP32 (Digital & Audio)

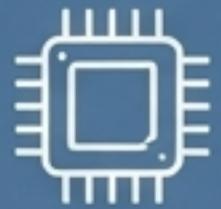


STM32 (Motion & Power)



Mode Validasi Khusus:
DAC vs PWM Compare

Target Output & Kebutuhan Sistem



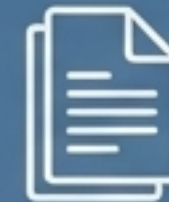
Target Hardware

- Speaker memproduksi output $\geq 85\text{dB}$.
- Pembacaan gelombang via Osiloskop.
- Respons aktuator mekanis linier (Servo & Motor).
- LED RGB tanpa flicker.



Arsitektur Software

- Struktur berbasis C++ Modular.
- Handler UART khusus (STM32/ESP32).
- Mekanisme logging data berformat CSV.



Dokumentasi Produksi

- Blueprint Arsitektur Sistem.
- Skematik Elektronik murni.
- Flowchart FSM presisi tinggi.
- Test Report komprehensif (Siap Production).

Rubrik Penilaian Teknis Proyek (20%)

Kriteria Standar: $A \geq 85$ | $B = 70-84$



DAC Waveform Smooth
(Frekuensi $\pm 1\%$)



Transisi Warna LED
(No Flicker)



Servo Motor Linier
(0-180°)



Tempo Buzzer
Presisi



Latency UART
Sinkron (<50ms)



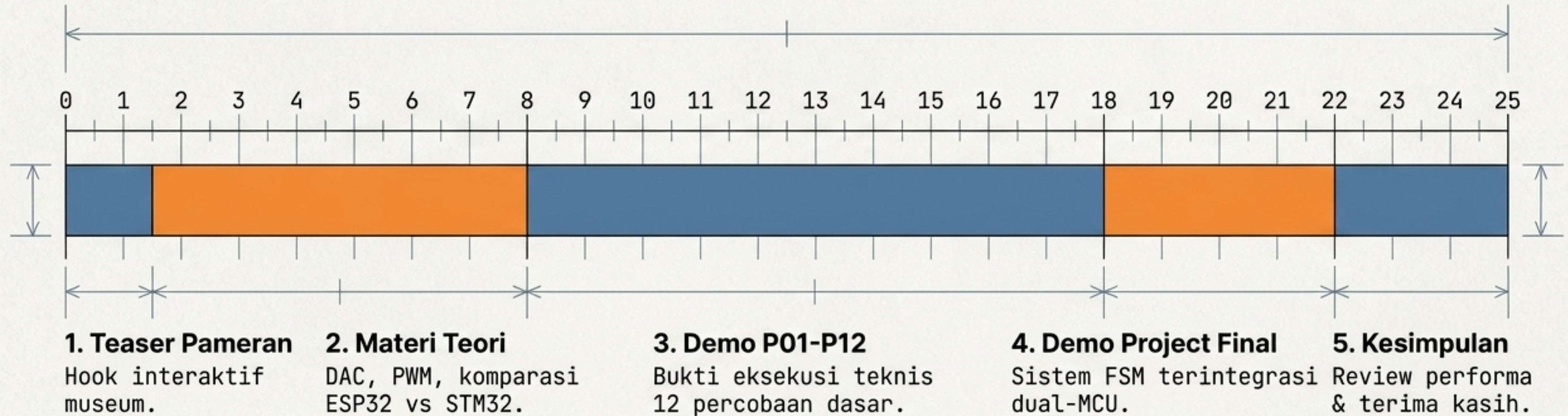
UX Mode Operasi
Instant



Kode C++ Modular

Struktur Video Dokumentasi

Durasi Wajib: 15-25 Menit | Format: YouTube Unlisted



Standar Eksekusi Rekaman Layar (PIP)

Integrasi Analisis P01 hingga P06

Callout 1

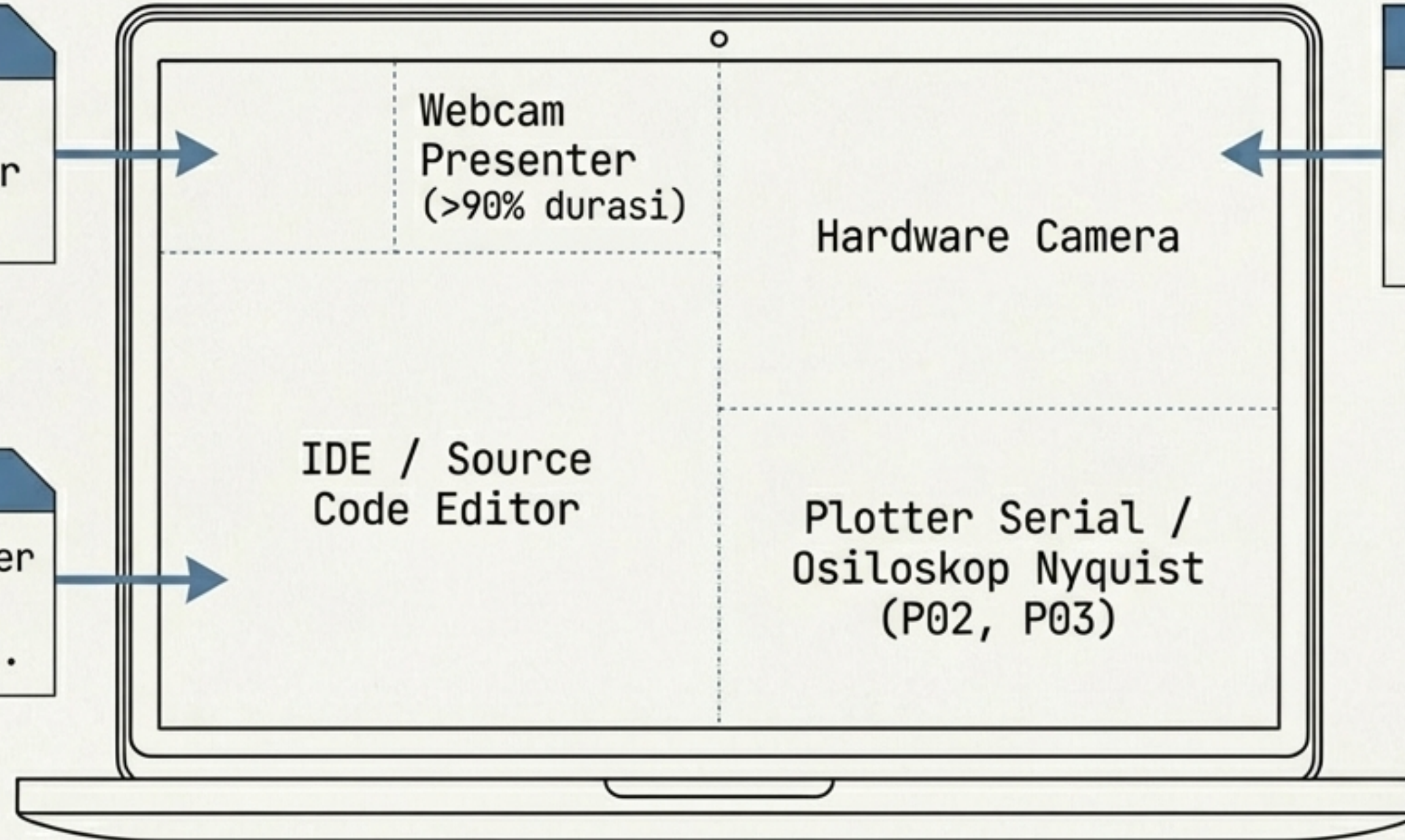
P01: Tampilkan layar Multimeter baca tegangan.

Callout 2

P04: Audio speaker terdengar jelas (pitch 440→880).

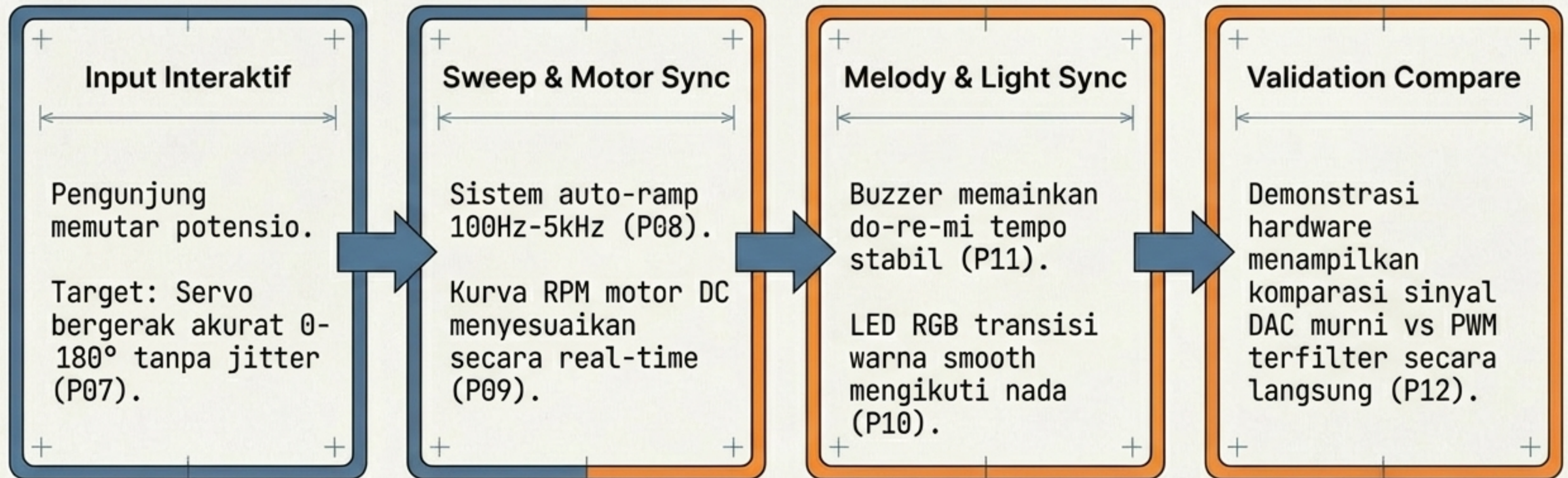
Callout 3

P05/P06: LED breathing smooth & real-time gamma plot.



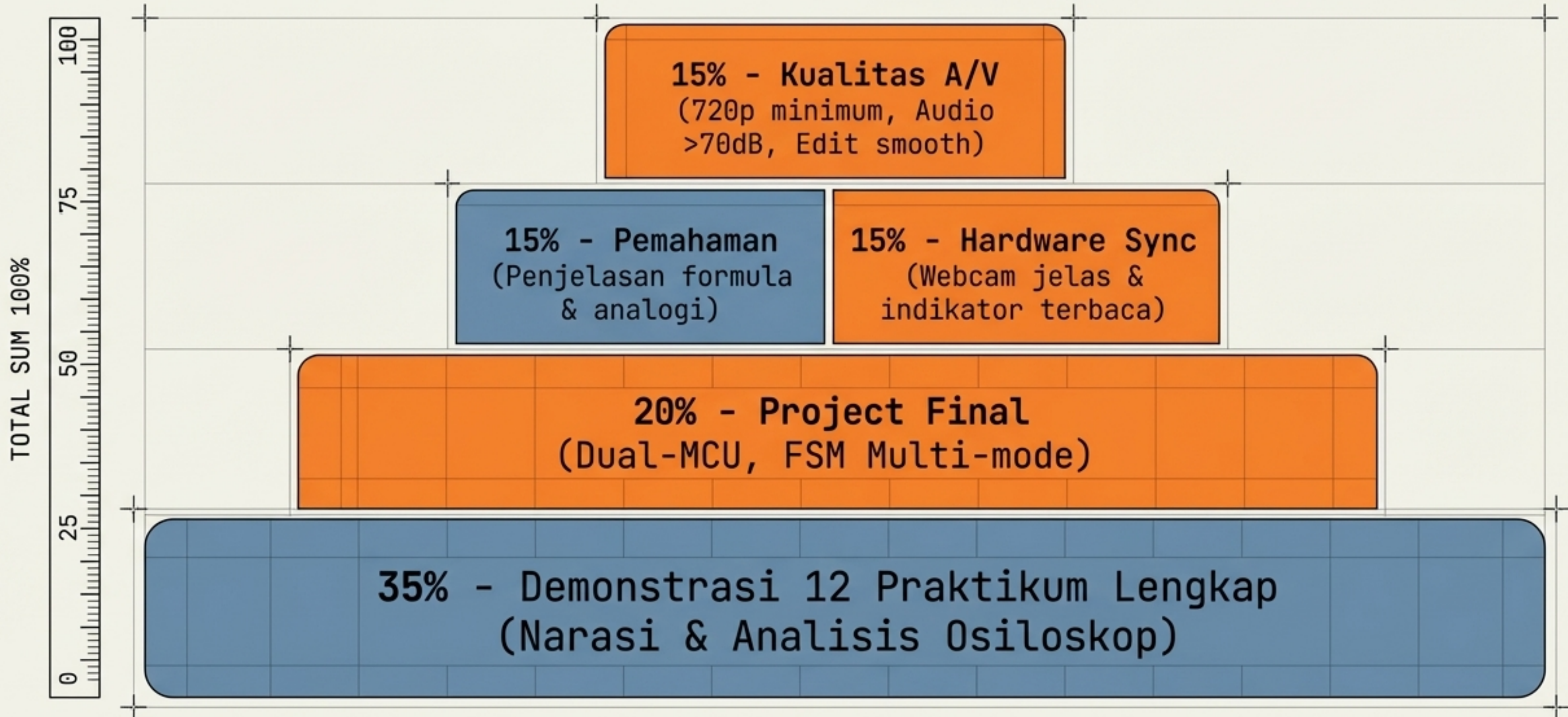
Demonstrasi Akhir: P07-P12 Final

Skenario: Panel Pengunjung Museum Sains



Komponen Penilaian Akhir

Skala: A $\geq$ 85 | B = 70-84 | C = 60-69 | D = 50-59



Checklist Final & Peringatan Penalti

Batas Akhir Submission via E-Learning

Pre-Flight Checklist

- ☒ Durasi di rentang 15-25 menit.
- ☒ Wajah di webcam >90%.
- ☒ Demo lengkap P01 hingga P12.
- ☒ Hardware & Project berfungsi penuh.
- ☒ Audio penjelasan jernih.
- ☒ Visual oscilloscope/serial terbaca.
- ☒ Source code dilampirkan.
- ☒ Link YouTube status Unlisted.

Penalti Fatal



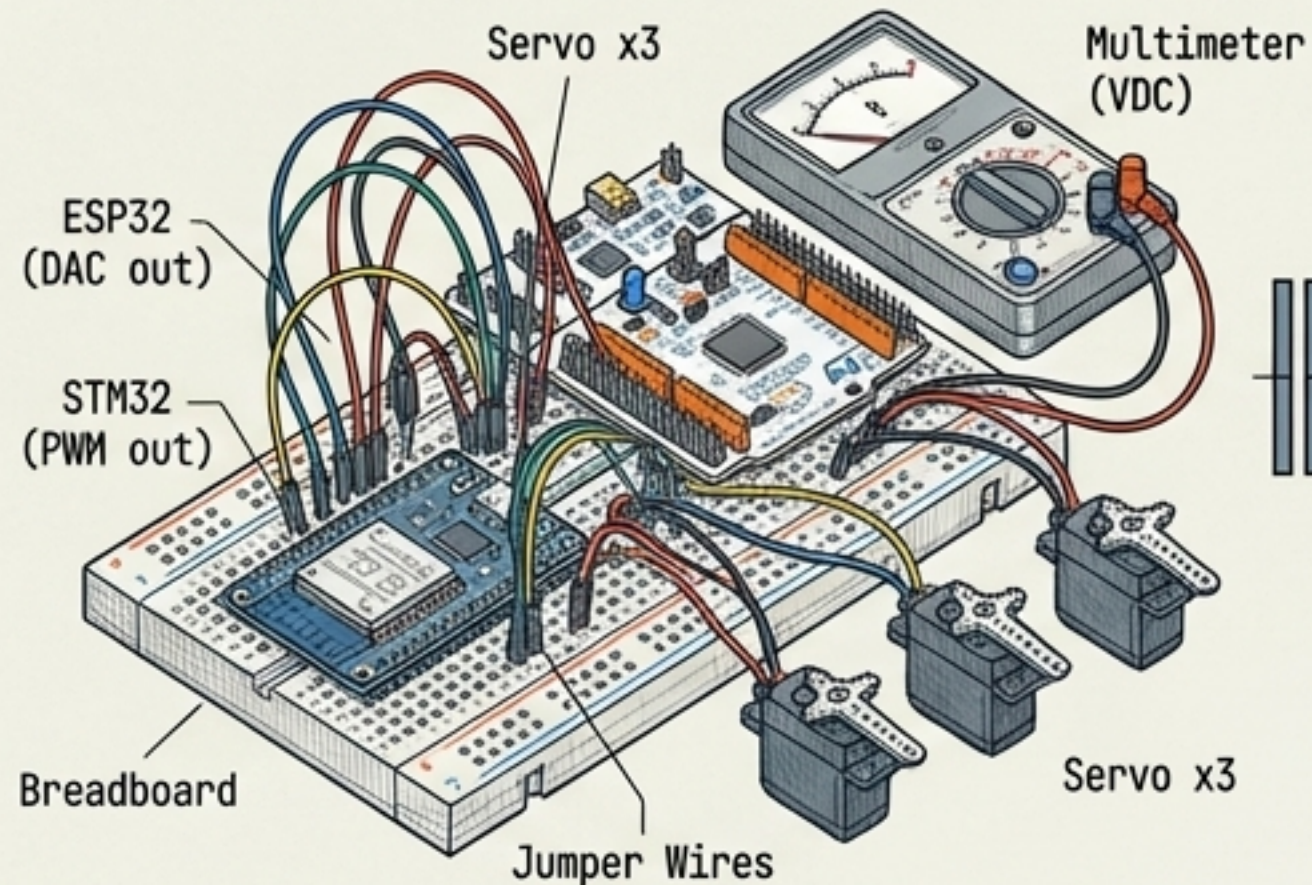
-  Incomplete Submission: -30%
-  Tidak Ada Hardware Terekam: -20%
-  Webcam Tidak Terlihat/Mati: -20%
-  Kualitas Audio Buruk: -15%
-  Resolusi Layar < 720p: -10%
-  Durasi < 10 Menit: -10%
-  Durasi > 30 Menit: -5%

Kendala Teknis? Hubungi TA maksimal 24 jam sebelum deadline via WhatsApp.

Kesimpulan Modul: Fondasi Industrial IoT

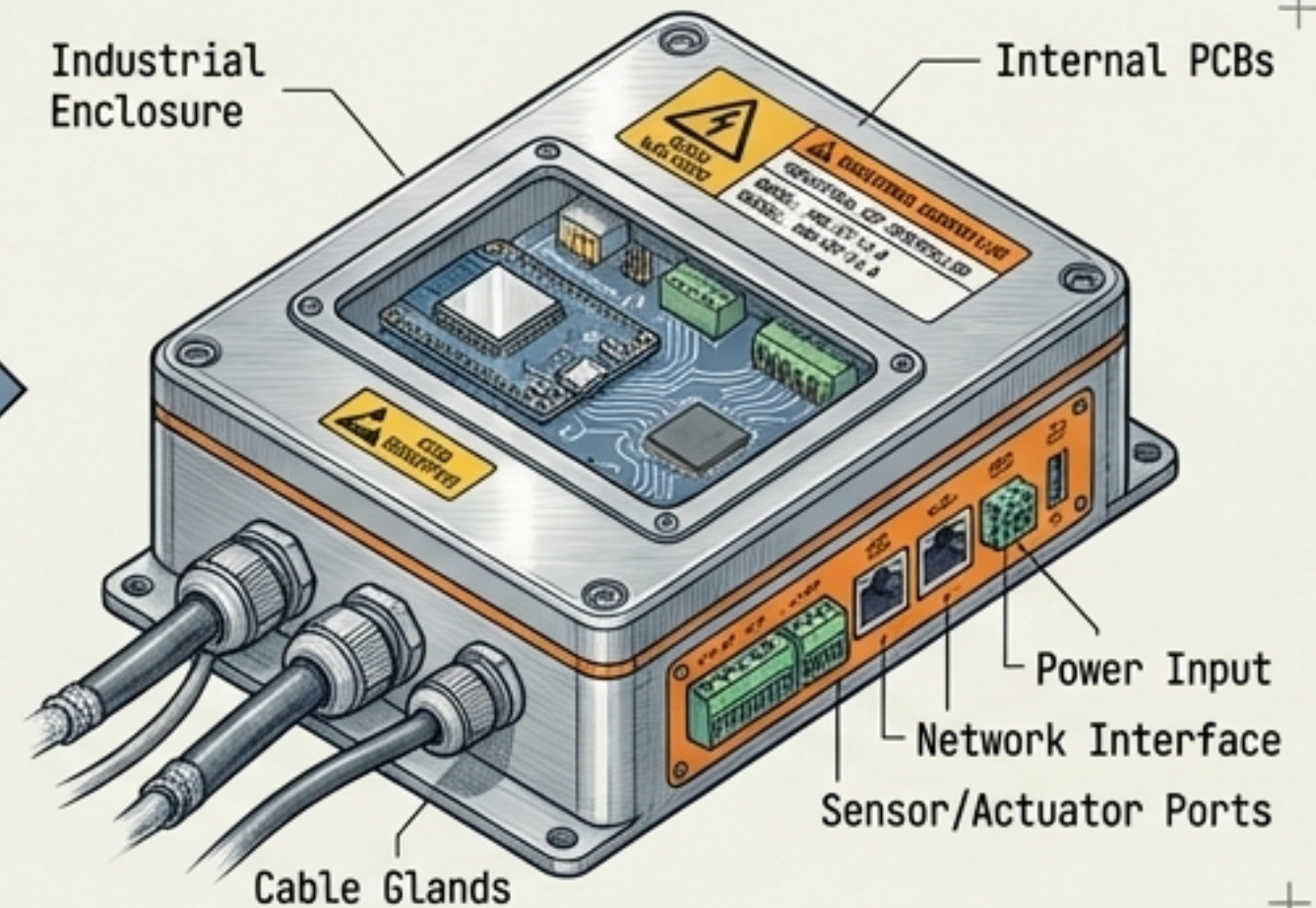
Dari Teori Sinyal ke Skala Deployment

+ Kompetensi Dasar (P01-P12):
Presisi DAC vs Efisiensi PWM



TRANSISI
ENGINEER

+ Industrial Scale Deployment:
Sinkronisasi Responsif Multi-MCU



Proyek ini melatih transisi dari hands-on prototyping menuju persiapan karir profesional di bidang Amplifier Audio, Motor Control, Smart Lighting, dan Embedded Systems.

Referensi Teknis & Toolchain

Standar Instrumentasi & Dokumentasi

Engineer's Workbench: Technical Reference & Toolchain Integration

Literatur Utama



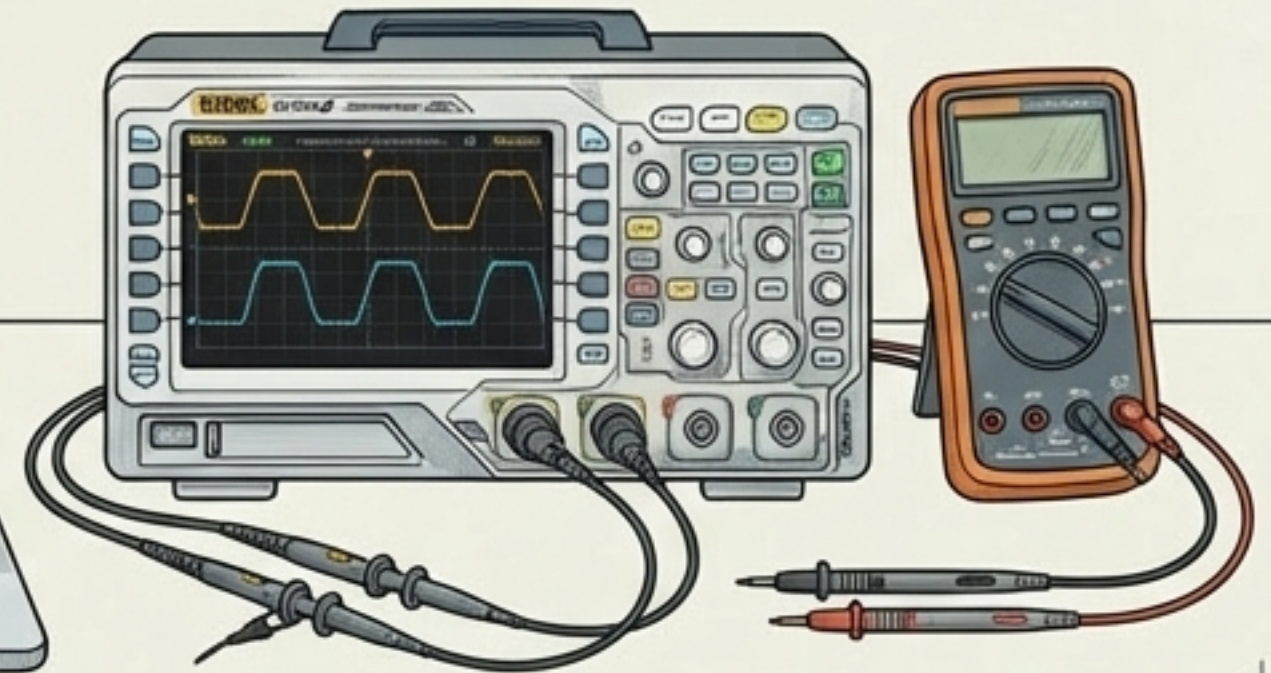
Development Tools

- VS Code & PlatformIO
- Python + Matplotlib (Data Plotting)
- Modular Reusable Codebase



Alat Ukur Instrumentasi

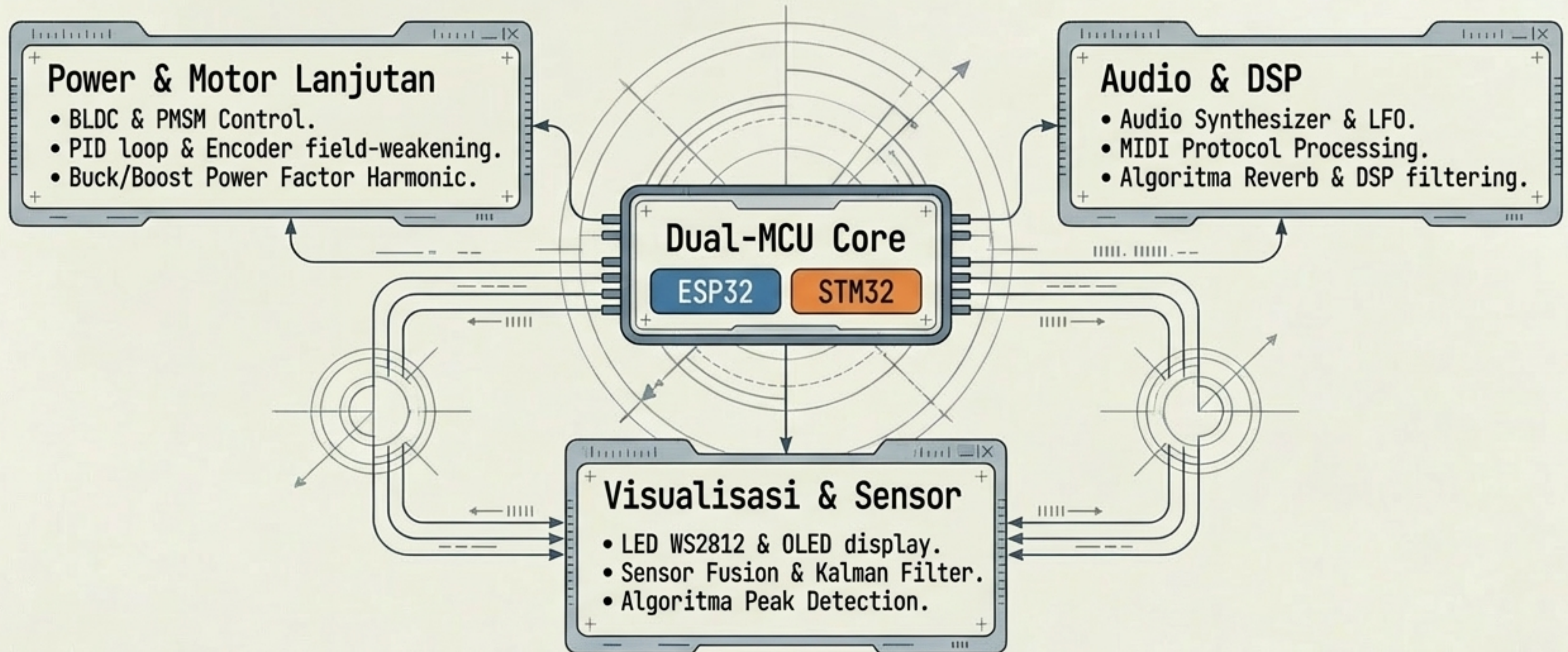
- Rigol DS1054Z Digital Oscilloscope
- Multimeter Digital Presisi



Dokumentasi dan standarisasi alat ukur serta kode merupakan fondasi untuk pengembangan sistem *embedded* yang andal dan dapat direplikasi secara konsisten.

Advanced Horizons: Eksplorasi Lanjutan

Skalabilitas Aktuator & Digital Signal Processing



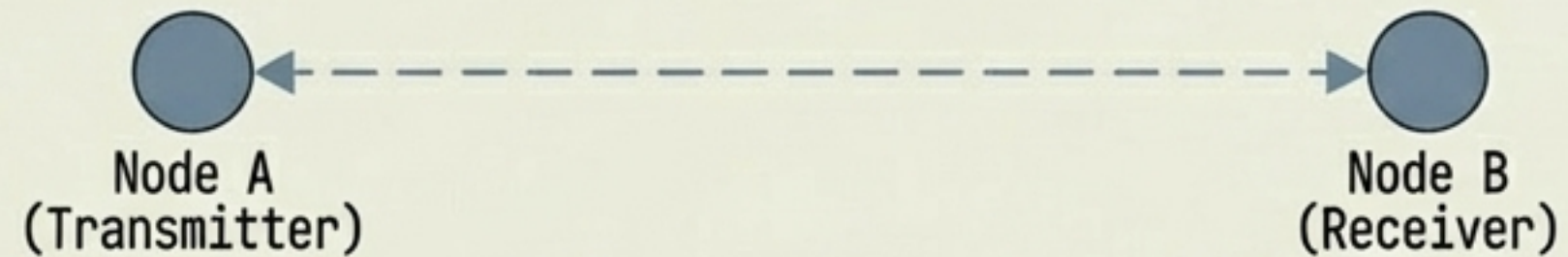
Sinkronisasi Kritis: Evolusi Jaringan Sistem

Dari Point-to-Point UART ke Industrial Bus

CURRENT STATE

Protokol Saat Ini: UART

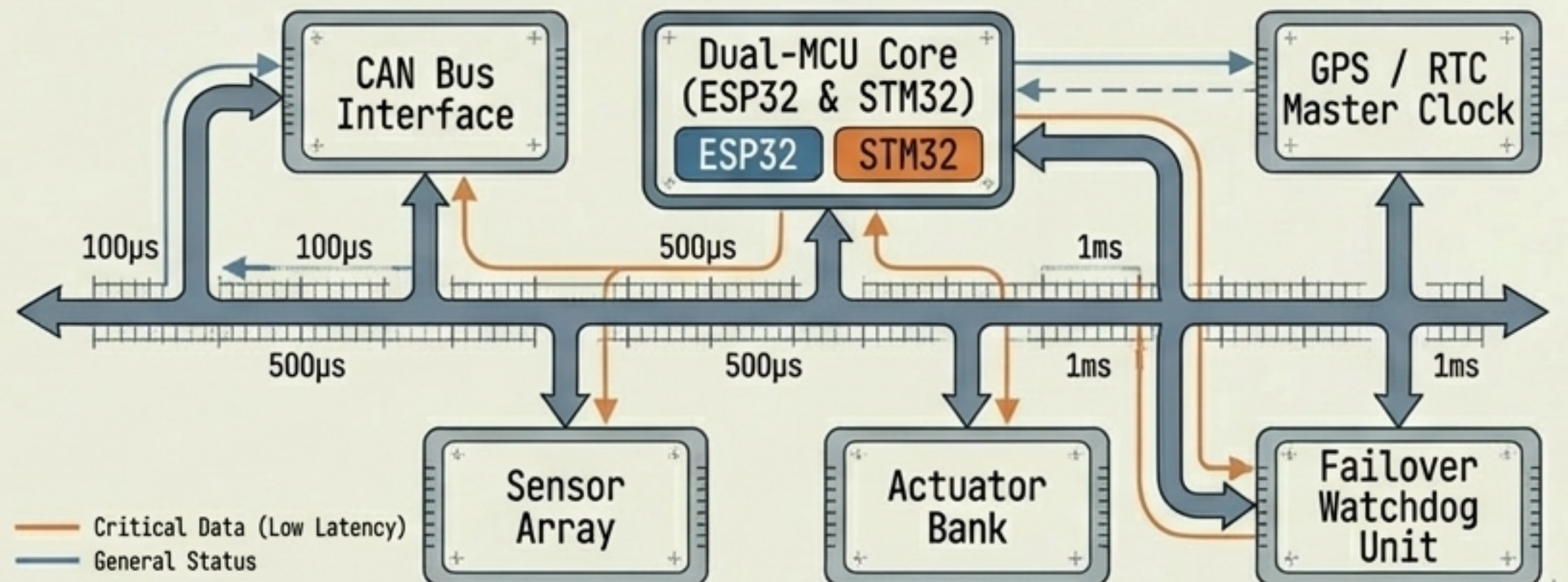
- Point-to-point sinkronisasi dasar.
- Latency & Jitter verifikasi via Logic Analyzer.



FUTURE STATE

Arsitektur Industri Skala Besar

- CAN Bus Arbitrasi (Prioritas data terjamin).
- FreeRTOS: Real-time event-driven (Microsecond precision).
- Sinkronisasi GPS / RTC Shared Network.
- Failover Watchdog untuk Safety Critical Systems.



Excellence in Engineering

Akhir Modul 05: DAC, PWM, dan Sistem Integrasi

Status Sistem: Kompeten.

- Teori, praktikum, proyek, dan dokumentasi video selesai.
- Konfigurasi DAC, PWM, waveform generator, aktuator servo, dan multi-MCU terkalibrasi.
- *Troubleshooting noise*, RC filtering, dan debugging osiloskop tervalidasi.
- Pemahaman *trade-off cost vs performance* (DAC vs PWM) tercapai.

Praktik yang konsisten menjawab tantangan industri. Siap melanjutkan ke penguasaan FreeRTOS, arsitektur daya tinggi, dan implementasi Wireless IoT.